Table S2. Frequency-sensitive exploratory candidates

Gene	Dataset	Frequency source	IVs	OR (95% CI)	*P* value	Q-test *P*
C4B	FinnGen GD	1000G EUR	2	1.63 (1.31–2.03)	1.10×10^{-5}	7.79×10^{-7}
C4B	FinnGen GD	eQTLGen cohort	2	1.75 (1.31–2.33)	1.48×10^{-4}	1.36×10^{-8}
C4B	FinnGen GO strict	1000G EUR	2	1.68 (1.44–1.95)	2.74×10^{-11}	0.122
C4B	FinnGen GO strict	eQTLGen cohort	2	1.80 (1.46–2.23)	6.17×10^{-8}	0.0587
HLA-G	FinnGen GD	1000G EUR	16	0.86 (0.72–1.03)	0.107	9.98×10^{-24}
HLA-G	FinnGen GD	eQTLGen cohort	16	0.60 (0.47–0.75)	1.72×10^{-5}	2.26×10^{-10}
HLA-G	FinnGen GO strict	1000G EUR	16	0.91 (0.76–1.08)	0.267	0.0231
HLA-G	FinnGen GO strict	eQTLGen cohort	16	0.70 (0.53–0.92)	0.0122	0.131

Exposure effects were reconstructed with 1000 Genomes Phase 3 European reference allele frequencies (1000G EUR; primary) or eQTLGen cohort frequencies; variant sets and outcome effects were held fixed, so changes reflect rescaled expression effects and IVW weights. *HLA-G* was added because its eQTLGen-frequency *P* crossed the screen threshold ($P < 0.05/2{,}544$). Q tests heterogeneity, not directional pleiotropy; its *P* value is not a gene-association *P* value. All 72 omission estimates and instrument weights are in the workbook. CI, confidence interval; IV, instrumental variable; IVW, inverse-variance weighted; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.